EK 301 Engineering Mechanics I

Final Design Report

Spring 2026

Boston University — College of Engineering

Group Members:

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Section: A2

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1 Introduction

The goal of this project was to design a truss that maximizes the load-to-cost ratio while meeting all specified constraints. This required finding a design capable of supporting a minimum load of 32 oz for 60 seconds at the lowest possible material cost. The design was subject to several constraints: the truss must be a simple truss composed of triangular units with straight members connected only at their ends, with no crossing or doubled members. The truss is supported by one pin and one roller support at either end, and the construction material was limited to two 4-foot and one 2-foot acrylic bars.

Our final design is a Howe-Warren hybrid configuration. A vertical member connects the load joint to the top chord, providing direct load transfer and local stiffness at the point of application. The remaining web members follow a Warren-style zigzag pattern, which distributes the applied force from the single load point across the full span of the truss efficiently and at low material cost. The tall profile of the top chord reduces internal member forces for a given applied load, improving buckling resistance across compression members. Together, these geometric choices allowed us to achieve a high predicted failure load relative to the total truss cost.

2 Procedure

Since the preliminary design report, a truss design was chosen and a small length modification was made to the design. In the preliminary design report, two trusses were designed– design 1, a Pratt truss with 9 joints and 15 members and design 2, a Howe-Warren truss with 8 joints and 13 members.

When evaluating the two designs against each other to conclude which one to proceed with building, the specifications were reviewed which revealed that design 1 had a member that was out of range. Thus, we proceeded with design 2, which fit the constraints. While design 2 was 4% percent more expensive than design 1, it had a higher load capacity and safety margin.

Other than finalizing the design to construct, no changes were made to the design procedure.

3 Analysis

The analysis procedure remained largely the same as in the preliminary design report. Member forces were determined using the method of joints, and buckling strengths were calculated using the power law fit formula:

$$F(L) = C\left(\frac{L_0}{L}\right)^2 \pm U_{\text{fit}} \tag{1}$$

where $C = 37.5$ oz, $L_0 = 10$ in, and $U_{\text{fit}} = 10$ oz. The nominal buckling strength $P_{\text{nom}} =$

$C(L_0/L)^2$ represents the predicted buckling load for a member of length L, with U_{fit} defining the uncertainty bounds above and below this value.

The one change made to the design was increasing member m7 from 5.00 in to 6.00 in to satisfy the minimum joint-to-joint span requirement of 6.00 in. This adjustment slightly increased the total member length and truss cost but did not change the critical member or significantly affect the predicted failure load.

To account for the uncertainty in the predicted failure load, three cases were analyzed, considering only compression members:

- **Nominal case:** each member's buckling strength $= P_{\text{nom}}$
- **Strong case:** each member's buckling strength $= P_{\text{nom}} + U_{\text{fit}}$
- **Weak case:** each member's buckling strength $= P_{\text{nom}} - U_{\text{fit}}$

The failure load was computed for each case through our software, yielding a predicted maximum load of 61.51 ± 11.95 oz, where the uncertainty ΔW was taken as the difference between the nominal and weak-case loads.

4 Results

The predicted maximum load of 61.51 ± 11.95 oz exceeds the 32 oz requirement at both the nominal and strong-case bounds. Member m4 (J1–J5, 7.81 in) was identified as the critical member, governing the failure load across all three cases.

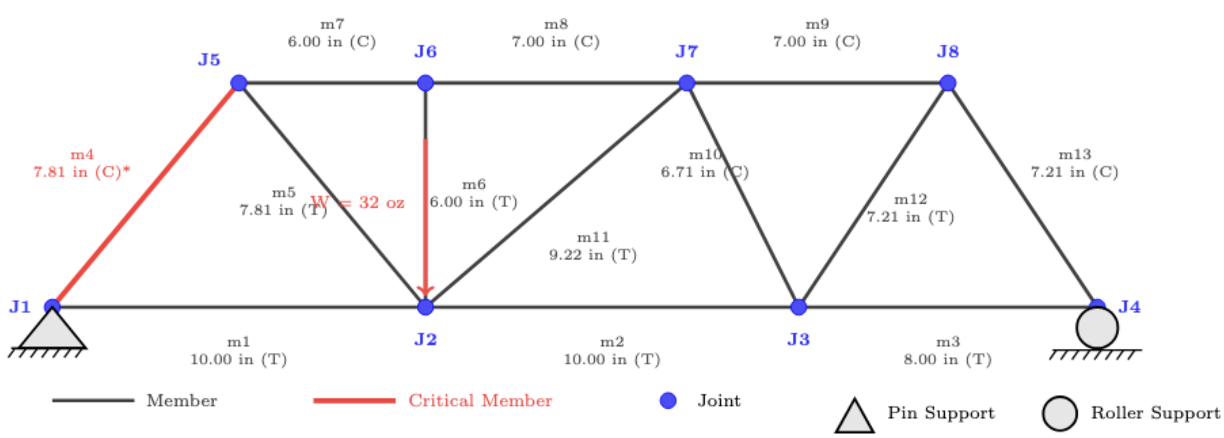

Figure 1: Design 2: Howe-Warren Hybrid Truss (8 Joints, 13 Members). Member lengths reflect the final design with m7 updated from 5.00 in to 6.00 in. T = tension, C = compression. (*) Critical member m4, predicted to buckle first.

Table 1: Member force summary for final design.

Member	Length (in)	T/C	Buckling Strength (oz)	Internal Force (oz)
m1	10.00	T	—	17.14
m2	10.00	T	—	20.95
m3	8.00	T	—	7.62
m4*	7.81	C	61.48 ± 10	26.78
m5	7.81	T	—	26.78
m6	6.00	T	—	0.00
m7	6.00	C	150.00 ± 10	34.29
m8	7.00	C	76.53 ± 10	34.29
m9	7.00	C	76.53 ± 10	15.24
m10	6.71	C	83.33 ± 10	12.78
m11	9.22	T	—	17.56
m12	7.21	T	—	13.74
m13	7.21	C	72.12 ± 10	13.74

*Critical member (predicted to buckle first).

Table 2: Design summary.

Property	**Value**
Total member length	99.97 in
Number of joints	8
Number of members	13 ($2J - 3 = 13$, simple truss: true)
Truss cost	$179.97
Critical member	m4 (length 7.81 in)
Maximum load	61.51 ± 11.95 oz
Load/cost ratio	0.3418 oz/$

5 Discussion & Conclusion

The final design was picked because it had the benefits of both Howe and Warren trusses, while also maximizing load-to-cost ratio and compliance with constraints. There is the Warren zig-zag pattern, which spreads load across the span efficiently. It is not a total Warren, however, because the vertical load joint (taken from the Howe) reduces local stress. It is better than just the Howe, because it reduces the total number of joints and therefore also cost. Optimizing the design consisted of trying to maximize the compression members resistance to buckling, which meant trying to employ a taller design. This could be seen in m4, which is the shortest compression member with the highest load–an intentional geometric choice. Such a choice was supported by the first lab, whose testing showed that the critical buckling force of acrylic strips was an inverse function of their length.

The most insightful and useful element turned out to be multi-verification cases with the aid of computer programs. Being able to develop and test without having to actually build different trusses of varying complexity provided invaluable data that helped inform the final design. Computer programs saved time and money while not conceding on accuracy.

In the future, if this project were to be done again, a change would be to have increased collaboration and information sharing between teams. While summary data such as Best Performance Truss for Each Section was made available, additional information such as the specific designs would be helpful as it would establish a shared baseline and raise the benchmark. If the best performing truss was provided to the whole class, it would then become the minimum, and so teams would be forced to revise their designs in an attempt to surpass the best performing truss. This would lead to more optimized designs overall.

A Meeting Minutes: Hartford Civic Center Collapse Discussion

Date:	April 22, 2026, 7:00 PM
Location:	ENG 202, Boston University, Boston, MA
Participants:	Ashley Dhar, Bilqees D'Urso, Jiahui Chen
Chair:	Ashley Dhar
Minute-Taker:	Bilqees D'Urso

Planned Agenda

1. Review of the 1978 Hartford Civic Center roof collapse
2. Discussion of computer programs in structural analysis and design
3. Determination of an appropriate safety factor for our truss if human life were at risk
4. Action items

Discussion Summary

- *Hartford collapse overview:* roof collapse occurred on January 18, 1978, shortly after the arena had been evacuated following a college basketball game. The main causes were three major design errors: underestimation of dead load, poor buckling resistance due to cruciform cross-sections, and failure to act on repeated deflection warnings during construction. (Jiahui)
- *Use of computer programs:* engineers relied heavily on computer analysis, which gave them a false sense of security; software did not account for buckling as a failure mode, so many vulnerabilities went undetected. Computer output should always be cross-checked with hand calculations and physical intuition. Even when the program flagged excessive deflection, engineers dismissed it - this should have been a warning sign. (Ashley)
- *Safety factors for our truss:* the Hartford roof frame showed excessive deflections at several nodes during ground inspection before erection. For our truss, we should examine the acrylic material beforehand to check for existing breaks or deformities. (Bilqees)
- *Safety factor for human life:* applying a safety factor requires accounting for material variability, construction imperfections, and uncertainty in loading—all demonstrated in the Hartford case. We agreed to apply the safety factor to the weak-case predicted failure load of 49.56 oz (i.e., $61.51 - 11.95$ oz), representing the most conservative realistic scenario. A safety factor of 2.0 is commonly used for structures where failure could endanger human life, yielding a claimed maximum load of $49.56/2.0 \approx 24.8$ oz for a life-safety application. Real structures would require additional consideration of dynamic loads, fatigue, and long-term material degradation not captured in our model. (Ashley)

- The material used at Hartford (some of the steel) did not meet specifications.

Conclusions

- Computer analysis is a valuable tool, but it does not substitute for engineering judgment or physical verification.
- Warning signs during construction should always be a cause for re-evaluation.
- In trying to cut costs and be innovative, the engineers did not sufficiently test their new idea.
- If our truss were used in a life-safety application, a safety factor of 2.0 applied to the weak-case failure loads yields a claimed maximum load of approximately 24.8 oz.